Cluster 4: Fieldwork

1 A group of four students wanted to investigate the effectiveness of the Beverage Container Return Scheme (BCRS) in encouraging recycling habits amongst residents in Bishan. The BCRS is a national recycling programme where used plastic beverage containers can be returned to reverse vending machines, known as "Return Right" machines, in exchange for a 10-cent deposit refund. The scheme aims to reduce plastic waste and support climate change mitigation efforts.

The hypothesis for their investigation was:

"The greater the convenience of recycling, the more likely that residents in Bishan will engage in recycling behaviours."

After secondary research, the students decided to conduct their investigation at Block 512 in Bishan, where a "Return Right" reverse vending machine is in front of a supermarket, and neighbouring blocks. They conducted their data collection on a Friday afternoon between 3:00 pm and 5:00 pm.

One student was stationed near the reverse vending machine at Block 512 for the 2-hour duration to conduct a pedestrian count and observe the utilisation of the reverse vending machine.

The remaining three students positioned themselves separately at various locations in the vicinity of the reverse vending machine and used convenience sampling to approach passers-by to complete a questionnaire survey. The questionnaire focused on their use of the reverse vending machine and their recycling habits. At the end of the 2-hour period, they had surveyed a total of 48 participants. Some of the students faced language barriers when communicating with elderly participants who spoke mainly in dialect.

Resource 1A shows a map of the investigation site and its vicinity. Resource 1B shows photographs of the investigation site and its vicinity. Resource 2 shows the profile of the respondents for the questionnaire survey. Resource 3 shows an extract from the questionnaire survey designed by the students. Resource 4A shows the results of Question 2 of the questionnaire survey. Resource 4B shows the results of the pedestrian count and observations of the utilisation of the Return Right reverse vending machine.

(a) Explain the strengths and limitations of the students' hypothesis. [5]

Award 1 mark for each explanation of a strength or limitation.
Award a maximum of 1 additional mark for further development of each explanation.
Award a maximum of 2 marks for explanation of either strength or limitation only.

Possible responses include:

Strengths:

- It is researchable (or measurable) as data on recycling behaviours and convenience of recycling can be collected through questionnaire surveys and observation of the reverse vending machine [1 mark]. The students can collect data on both variables in the hypothesis through primary data collection methods. [1 additional mark]

- It is at an appropriate spatial scale as the investigation is focused on Bishan, which is a singular site and therefore manageable for a group of four students to conduct their fieldwork within a 2-hour period [1 mark].

Limitations:

- It is not clearly defined as "convenience of recycling" is not specific and could refer to multiple aspects such as proximity to the reverse vending machine, opening hours, ease of use, or awareness of the scheme [1 mark]. This means the students may not be able to design their data collection methods to capture the appropriate information needed, as it is unclear which aspect of convenience they are investigating [1 additional mark].

(b) With reference to Resources 1A and 1B, suggest how the students could minimise potential risks and impacts during their data collection. [6]

Award 1 mark for each suggestion on how the students could minimise potential risks and impacts during data collection, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

Minimisation of risks:

- Students should work in pairs rather than individually at each survey location [1 mark]. As seen in Resource 1A, the three students conducting questionnaire surveys are each stationed alone at Blocks 510, 511, and 513, which means there is no one to assist if a student encounters a difficult or threatening situation [1 additional mark].

- Students should position themselves to the side of walkways and avoid standing in the middle of high-footfall areas when conducting questionnaire surveys [1 mark]. As seen in Resource 1B, the survey site at Block 511 is in a busy coffeeshop area where narrow passageways could obstruct pedestrian flow and increase the chances of collisions with other users of the coffeeshop, resulting in injuries to the students. [1 additional mark]

Minimisation of impacts:

- Students should keep their voices at a low volume and be mindful of noise levels when conducting questionnaire surveys [1 mark]. The survey site at Block 511 is located near a hawker centre as seen in Resource 1B, where residents may be dining and could be disturbed by prolonged conversations [1 additional mark].

- Students should be respectful and not loiter or crowd around the "Return Right" reverse vending machine at Block 512 when conducting the pedestrian count [1 mark]. As seen in Resource 1B, the machine is located in front of a Sheng Siong supermarket entrance, and obstruction of the area could inconvenience shoppers and affect the normal operation of the machine [1 additional mark].

(c) Suggest how the students could ensure that their investigation is conducted ethically. [4]

Award 1 mark for each suggestion on how the students could ensure their investigation is conducted ethically, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

- Students should inform respondents of the purpose of the investigation before administering the questionnaire survey and obtain their consent before proceeding [1 mark]. This ensures that participation is voluntary and that respondents are aware of how the information they provide will be used, protecting their rights as participants [1 additional mark].

- Students should not collect any personally identifiable information such as full names, home addresses, or identification numbers in the questionnaire [1 mark]. This protects the privacy and anonymity of respondents, ensuring that their personal data cannot be traced back to them [1 additional mark].

(d) With reference to Resource 2, explain why the sample for the questionnaire survey may not be representative of the profile of residents in Bishan. [4]

Award 1 mark for each explanation of why the sample may not be representative, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- The sample may not be representative of the age profile of residents in Bishan as a disproportionately large number of respondents are older or elderly. [1 mark]
    - As seen in Resource 2, 56.2% of respondents are aged 56 and above, suggesting an over-representation of older and elderly residents [1 additional mark].
    - This is likely because data collection was conducted on a Friday afternoon between 3:00 pm and 5:00 pm, a time when working-age adults are less likely to be present in the area, skewing the sample towards older residents who are more likely to be at home or in the vicinity during that time [1 additional mark].
- The sample may not be representative of the gender profile of residents in Bishan as a disproportionately large number of respondents are female. [1 mark]
    - As seen in Resource 2, 60.4% of respondents are female, suggesting an over-representation of female residents [1 additional mark].

(e) Explain the strengths and limitations of the design of the questionnaire survey shown in Resource 3. [6]

Award 1 mark for each explanation of a strength or limitation of the questionnaire design, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.
Award a maximum of 3 marks for explanation of either strength or limitation only.

Possible responses include:

Strengths:

- The questionnaire uses a mix of close-ended and open-ended questions, allowing for the collection of both quantitative and qualitative data that enhances comprehensiveness of data collected [1 mark].
    - Close-ended questions such as Questions 1 to 3 allow for ease of comparison and analysis of trends in recycling habits across respondents, while the open-ended Question 4 allows for more in-depth understanding of the factors that would encourage recycling, contributing to a more comprehensive dataset [1 additional mark].

Limitations:

- Some response options in the questionnaire are subjective and open to interpretation, which reduces the reliability of the data collected. [1 mark]
    - The frequency options in Question 2 such as "all the time", "very often", "often", and "sometimes" are not defined by specific numerical thresholds. Different respondents may interpret these terms differently, making it difficult to draw consistent and reliable conclusions about usage patterns [1 additional mark].
- Some questions only offer binary response options to capture the full range of respondents' views, reducing the reliability of the data collected. [1 mark]
    - Question 3 asks whether the reverse vending machine is convenient to use but only provides "yes" and "no" as options. This binary format forces respondents to take an extreme position even if their view is more nuanced or neutral, and hence their responses may not accurately reflect their views [1 additional mark].

(f) With the aid of a sketch diagram, explain how the students could better represent the data shown in Resource 4A. [5]

Award 1 mark for an appropriate choice of data representation method.
Award 1 mark for a sketch diagram that accurately reflects the data in Resource 4B.
Award 1 mark for appropriate labelling of the sketch diagram, including a title, axis labels or segment labels, and units where applicable.
Award up to 2 marks for explanation of why the chosen method better represents the data.

Possible responses include:

- A bar graph or pie chart allows for immediate visual comparison of the distribution of responses, making it easier to identify and analyse trends [1 mark].
    - This is more effective than a table, where the reader must actively compare numerical values across rows, making it harder to identify patterns and trends at a glance [1 additional mark].

(g) The students concluded that greater convenience does not increase in the likelihood of engaging in recycling behaviours amongst residents in Bishan. Evaluate the validity of their conclusion. [10]

Possible Approaches

Candidates could agree that the conclusion may be valid as the **results from the investigation support the conclusion made**. Resource 4A shows that only 15.4% of people who interacted with the reverse vending machine successfully utilised it to return used bottles, suggesting that the convenience of having the machine located just outside a supermarket did not translate into greater use of the machine and hence did not improve recycling behaviours. This is further supported by the observation in Resource 4A that 76.9% of people approached the machine but walked away without using it. Resource 4B also corroborates this, as 54.2% of survey respondents (26 out of 48) indicated that they had never used the machine, suggesting that proximity to the machine did not encourage recycling habits amongst residents.

Candidates could critique the **limited reliability of the investigation due to the unrepresentative sample profile** shown in Resource 2. As 56.2% of respondents are aged 56 and above and 60.4% are female, the sample is unlikely to reflect the actual age and gender distribution of residents in Bishan. This limits the reliability of the questionnaire results and hence the validity of the conclusion drawn.

Candidates could also critique the **limited reliability of the data collected due to weaknesses in the questionnaire design** shown in Resource 3. The use of ambiguous frequency terms such as "very often", "often", and "sometimes" in Question 2 allows for subjective interpretation by different respondents. As students' and respondents' understanding of the terms may differ, this may lead to inaccurate interpretation of the results and the data collected may not reflect actual usage patterns. This reduces the reliability of the results and hence the validity of the conclusion.

Candidates could also question the **accuracy of the data collected** due to the language barriers described in the preamble. Some elderly participants communicated in dialect, which may have led to misunderstandings during the questionnaire survey. If responses were inaccurately recorded as a result, this would reduce the accuracy of the data collected and in turn limit the validity of the conclusion drawn.

Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions.

Section B

Cluster 3: Sustainable Future and Climate Change

2 Seoul, South Korea, unveiled its new brand slogan "Seoul, My Soul," in August 2023. Resource 5A is an article on the challenges of city branding efforts in Seoul, South Korea. Resource 5B provides information on the related initiatives of the "Seoul, My Soul" city branding campaign.

Resource 6 shows the living arrangements for seniors (65 years old and above) in resident households in 2015 and 2025. Resource 7 shows responses of Singaporeans to a survey in 2018 on the importance of four sectors in bearing responsibility for eldercare.

(a) Explain how urban population trends in both developing and developed countries challenge progress towards sustainable development. [6]

Award 1 mark for each explanation of how urban population trends challenge progress towards sustainable development, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Award a maximum of 3 marks for responses that address urban population trends in only one context.

Urban population trends in developing countries:

- High urbanisation rates and rapid urban growth in developing countries result in a rapidly increasing urban population that outpaces the provision of adequate infrastructure such as basic sanitation and access to clean water [1 mark], meaning that the basic needs of a growing urban population cannot be met, hindering social sustainability and progress towards sustainable development [1 additional mark].

- Rapid urban growth in developing countries leads to the expansion of urban slums, as the urban poor are forced to reside in informal settlements with inadequate housing and services [1 mark], deepening urban poverty and inequality and making it difficult for cities to improve the quality of life for all residents, thereby hindering social sustainability [1 additional mark].

- High urbanisation rates in developing countries are associated with a lack of formal sector employment opportunities, as the working-age population grows faster than gainful employment [1 mark], leading to the informalisation of the urban economy where workers are employed in unregulated and low-paying activities, deepening urban poverty and hindering economic sustainability [1 additional mark].

- Rapid urban growth in developing countries generates massive amounts of untreated and uncollected waste as waste management infrastructure fails to keep pace with population growth [1 mark], leading to the contamination of water sources and soils, which undermines

environmental sustainability and poses significant risks to the health of urban residents [1 additional mark].

Urban population trends in developed countries:

- Urban population loss in developed countries reduces the size of the highly skilled workforce contributing to the city's economy [1 mark], potentially limiting the resources available to the city for economic, social, and environmental development, and deterring investors due to a smaller market size, thereby hindering economic sustainability [1 additional mark].
- Urban population loss in developed countries is strongly correlated with the ageing of urban populations, as younger residents leave cities in search of better opportunities elsewhere [1 mark], placing greater pressure on health and long-term care systems and escalating costs that a shrinking tax base may be unable to meet, thereby hindering both economic and social sustainability [1 additional mark].

- Shrinking cities in developed countries are characterised by urban decay such as property abandonment and deteriorating infrastructure [1 mark], creating a vicious cycle where falling property values reduce tax revenues, making it harder for the city to fund public services and maintain living conditions, further driving population loss and hindering progress towards economic and social sustainability [1 additional mark].

- Urban population loss in developed countries can result in the abandonment of previously developed land and buildings [1 mark], leading to environmental degradation as neglected sites may become sources of pollution and contamination, and the costs of remediating such brownfield sites may be beyond the means of a city with a shrinking tax base, hindering environmental sustainability [1 additional mark].

(b) Explain how efforts to achieve sustainable urban development and enhance liveability of cities may be complementary. [4]

Award 1 mark for each explanation of how efforts to achieve sustainable urban development and enhance liveability may be complementary, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses:

- Liveability initiatives can serve as incremental steps that collectively contribute to the achievement of sustainable urban development in the long term [1 mark].
    - For example, efforts to improve walkability or expand public green spaces in a city enhance the immediate quality of life for residents, while also reducing carbon emissions and mitigating the urban heat island effect over time, thereby advancing broader sustainability goals [1 additional mark]. **OR**
    - For instance, urban greening initiatives that introduce parks and green corridors into the built environment provide residents with recreational spaces and improved physical and psychological health, thereby enhancing liveability but also support carbon sequestration and biodiversity conservation — key sustainability outcomes [1 additional mark].

- Liveability helps to make the abstract vision of sustainable urban development more tangible and relevant to residents and decision-makers at the local level [1 mark].
    - Since liveability focuses on immediate and localised improvements to the urban environment — such as improving public transport or reducing noise pollution — it provides a practical pathway through which sustainability goals can be translated into concrete actions that people can experience in their daily lives [1 additional mark].

- Sustainable urban development provides the long-term vision and boundary conditions within which liveability initiatives must operate, while liveability provides the day-to-day, evolving path towards achieving that vision [1 mark].
    - In this sense, the two concepts are mutually reinforcing — sustainability ensures that liveability improvements do not come at the expense of future generations, while liveability ensures that sustainability goals remain grounded in the lived experiences and needs of present residents [1 additional mark].

(c) With reference to Resources 5A and 5B, suggest how Seoul's new city branding campaign and its related initiatives can help the city progress towards sustainable urban development. [6]

Award 1 mark for each suggestion of how Seoul's new city branding campaign and its related initiatives can help the city progress towards sustainable urban development, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Award maximum of 4 marks for making reference from either Resource 5A or 5B only.

Possible responses:

With reference to Resource 5A — the "Seoul, My Soul" campaign:

- Resource 5A states that the "Seoul, My Soul" campaign was developed through public consultation and is focused on local pride and well-being, which could strengthen residents' sense of ownership and trust in local governance [1 mark], supporting the **social dimension** of sustainable urban development by ensuring that citizens feel meaningfully involved in shaping their city [1 additional mark].

- The "Seoul, My Soul" campaign aims to enhance Seoul's urban competitiveness and attract visitors and investors by building a clearer and more accessible city identity [1 mark], which could help sustain and grow Seoul's urban **economy** by drawing in new investment and tourism revenue, supporting the economic dimension of sustainable urban development [1 additional mark].

- Resource 5A notes that the "Seoul, My Soul" campaign uses colourful and accessible symbols focused on well-being and local pride [1 mark], which could foster a stronger sense of community identity and belonging among residents, contributing to **social** sustainability by improving quality of life and reducing feelings of alienation in the city [1 additional mark].

With reference to Resource 5B — related initiatives:

- Resource 5B describes the Seoul Without Loneliness initiative, which establishes accessible neighbourhood spaces where residents can rest, connect, and access emotional support without pressure, directly addressing social isolation and improving the well-being of residents [1 mark], which supports the **social** dimension of sustainable urban development by improving quality of life and reducing inequality in access to mental health support [1 additional mark].

- The Seoul Mind Convenience Store described in Resource 5B provides professional counselling and peer support from trained volunteers who have experienced isolation themselves [1 mark], making mental health

support more accessible to a wider range of residents regardless of their socioeconomic background, thereby supporting **social** sustainability through more equitable access to community services [1 additional mark].

- Resource 5B describes the Garden City Seoul initiative, which aims to connect fragmented urban green spaces and make nature accessible to every citizen within a five-minute reach [1 mark], directly addressing **environmental** sustainability by integrating green spaces into everyday urban life, which helps mitigate the effects of climate change in the city 1 additional mark].

- The Garden City Seoul initiative described in Resource 5B integrates green spaces with leisure programmes that combat social isolation and youth seclusion, demonstrating that the initiative supports **social** sustainability through the reduction of isolation [1 mark].

(d) With reference to Resource 5A, explain why city branding efforts in Seoul may limit its progress towards sustainable urban development. [4]

Award 1 mark for each explanation of why city branding efforts in Seoul may limit its progress towards sustainable urban development, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible Responses

- The photograph in Resource 5A shows workers physically dismantling the large "I.SEOUL.U" letter installations along the riverbank, which suggests that significant amounts of public money and manpower are being spent on removing branding infrastructure rather than on more meaningful urban development projects that could improve residents' quality of life, hindering social and economic sustainability. [1 mark]

- According to Resource 5A, Seoul has gone through multiple rebranding exercises - from "Hi Seoul" to "I.SEOUL.U" and now to "Seoul, My Soul" - each time following a change in city leadership rather than being driven by long-term community planning [1 mark].
    - This means that resources are repeatedly spent on short-term, politically motivated decisions instead of being channelled into sustained urban development efforts [1 additional mark].

- Resource 5A states that residents have come to view public voting on city branding as a superficial exercise, which means that citizens feel their opinions do not genuinely shape urban decisions. [1 mark]
    - This weakens the sense of community ownership and trust in local government, making it harder to achieve the social dimension of sustainable urban development, which requires fair participation and access for all residents [1 additional mark].

- Resource 5A highlights that the constant cycle of rebranding has left global audiences confused and weakened Seoul's long-term international recognition, which could put off potential investors and tourists. [1 mark]
    - Without sustained economic investment, Seoul may struggle to fund the infrastructure and services needed to support sustainable urban development, and its goal of becoming a top five global city may be further out of reach [1 additional mark].

(e) With reference to Resource 6, describe how the living arrangements for seniors in Singapore have changed from 2015 to 2025. [5]

Award 1 mark for each accurate description of a change in the living arrangements of seniors in Singapore from 2015 to 2025, to a maximum of 5 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses:

- Resource 6 shows that the number of seniors **living alone** in households increased. [1 mark]
  - The number grew from 41,200 in 2015 to 88,400 in 2025/ an increase of approximately 47,200 seniors. [1 additional mark] **OR**
  - As a proportion of the total senior population, seniors living alone increased from 8.9% in 2015 to 11.5% in 2025/ an increase of 2.6%. [1 additional mark]

- Resource 6 shows that the number of seniors **living with a spouse but no children increased**. [1 mark]
  - The number grew from 107,300 in 2015 to 222,800 in 2025/ an increase of approximately 115,500 seniors. [1 additional mark] **OR**
  - As a proportion of the total senior population, seniors living with a spouse but no children increased from 23.3% in 2015 to 29% in 2025/ an increase of 5.7%. [1 additional mark]

- Resource 6 shows that the number of seniors **living without a spouse but with at least one child** increased. [1 mark]
  - The number grew from 116,500 in 2015 to 119,900 in 2025/ an increase of approximately 3,400 seniors. [1 additional mark]
- Resource 6 shows that the <u>proportion</u> of seniors **living without a spouse but with at least one child** <u>decreased</u>. [1 mark]
  - As a proportion of the total senior population, this category decreased from approximately 25.3% in 2015 to 15.6% in 2025/ a decrease of 9.7%. [1 additional mark]

- Resource 6 shows that the number of seniors **living with both a spouse <u>and</u> at least one child** increased. [1 mark]
  - The number grew from 166,000 in 2015 to 272,100 in 2025/ an increase of approximately 106,100 seniors. [1 additional mark]

- Resource 6 shows that the <u>proportion</u> of seniors **living with both a spouse and at least one child** <u>decreased</u>.
  - As a proportion of the total senior population, this category decreased from approximately 36.0% in 2015 to 35.4% in 2025/ a decrease of 0.6%. [1 additional mark]

(f) With reference to Resources 6 and 7, explain why it might be challenging to meet the needs of the elderly in Singapore. [5]

Award 1 mark for each explanation of why it might be challenging to meet the needs of the elderly in Singapore, to a maximum of 5 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses:

- Resource 6 shows that the total senior population grew significantly which suggests that there are more and more elderly people who need care. [1 mark].
    - As the number of elderly people grows rapidly, it becomes harder for families, the community, and the government to provide enough support and services for all of them. [1 additional mark]

- Resource 7 shows that 73.4% of Singaporeans rank family as the most important sector in bearing responsibility for eldercare which suggests that there might be disproportionately high expectations on the family to care for the elderly/ too much responsibility for caring for the elderly falls on the family. [1 mark]
    - This means that if an elderly person does not have family members who are able or willing to care for them, their needs may go unmet, as other sectors like employers and the community are not seen as equally responsible for stepping in. [1 additional mark]

- Resource 6 shows that there is an increasing proportion of elderly people living alone without family support, yet Resource 7 shows that family is still expected to be the main caregiver. [1 mark]
    - This is particularly challenging as for elderly living in such circumstances, their daily care and emotional needs may not be adequately met since family support is simply not available. [1 additional mark]

- Resource 6 shows that the proportion of seniors living without a spouse but with at least one child fell sharply from 25.3% to 15.6% / the proportion of seniors living with a spouse but no children increased from 23.3% to 29%, suggesting that fewer elderly people have children at home to help look after them. [1 mark]
    - This makes it harder to meet their physical and emotional needs on a daily basis, especially as they grow older and more dependent. [1 additional mark]

- Resource 7 shows that community is most commonly ranked third by 45.4% of respondents and employers are ranked last by 57.1%, suggesting that most Singaporeans do not see these sectors as having a major role in caring for the elderly. [1 mark]
    - This is a challenge because as more elderly people live alone or without family support — as shown by the growing number of

seniors living alone in Resource 6 — the community and employers are not stepping up sufficiently to fill the gap, leaving some elderly without adequate care. [1 additional mark]

Section C

Answer EITHER question 3 OR question 4.

3 "Cities today are sufficiently liveable for women."

Evaluate this statement. [20]

Possible Approaches:

Candidates could approach the question by making a judgement on the validity of the statement through a consideration of the issues currently faced by women in cities and the relative strengths and limitations of strategies implemented to improve the liveability of the city for women. Candidates could employ evaluative criteria to evaluate the extent and severity of issues related to economic, social and psychological well-being and the effectiveness of strategies to address these issues to improve liveability. Candidates could also consider strategies implemented at different spatial scales or by different stakeholders.

Candidates could also approach the question by making a judgement on whether some cities have been more successful in addressing issues faced by women in cities than others, through a consideration of two or more case studies. Candidates could evaluate whether the strategies in each case achieved their intended aims and analyse any unintended negative impacts. Candidates could also analyse contextual factors such as the scale of the issues, the political will of the authorities, cultural attitudes towards gender equality, and the resources available, which may have contributed to variations in success between places.

Levels marked using Generic Level Descriptors for H2 essays.

4 Evaluate the extent to which contemporary climate change can be attributed to natural factors. [20]

Possible Approaches:

Candidates could approach the question by making a judgement on whether contemporary climate change can be attributed to **natural factors** (including variations in solar output, the thermohaline circulation, volcanic activity, or the El Nino phenomenon) **or human factors** (such as the burning of fossil fuels or deforestation that lead to increased concentration of greenhouse gases). Candidates could consider evidence of past climate variability and episodes of warming and cooling of Earth during the Quaternary period, and how far they correspond with variations in natural factors or changes in human activities over a similar period.

Candidates could approach the question by making a judgement on the **relative extent** to which natural factors are responsible for contemporary climate change. Candidates could evaluate the significance of each natural factor by considering the scale and duration of its influence on global temperatures, and assess whether the patterns of contemporary climate change correspond with variations in these natural factors.

Levels marked using Generic Level Descriptors for 20m H2 essays.